

From Vols to Correlation

A Forward-Looking Framework for FX Correlation Trading and Cross Asset Portfolio Integration

Extel - 2025 Institutional Investor
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- We extend our FX vol surface forecasting framework to model implied correlation, leveraging the triangular no-arbitrage identity to construct a consistent and forecastable correlation surface across 41 FX pairs.
- The model focuses on ATM-only volatility inputs, **simplifying the surface structure to two term structure parameters while dropping smile-related components**. VAR model is then used to forecast surface coefficients. This combined parametric specification approach achieves strong in-sample and out-of-sample fit and interpretability.
- We conducted three systematic correlation trading strategies by comparing market-implied correlations against model-implied, against realized (Benchmark), and using their intersection (Hybrid). Our backtest has delivered robust performance since 2012. **The model-implied correlation provides a more robust and timely signal than relying on realized correlation alone, and the combined signal further enhances the risk-adjusted return, particularly in recent years, reinforcing its quality as a high conviction approach.**
- Beyond its standalone performance, this model meaningfully **enhances our cross-asset portfolio when integrated within our broader multi-strategy framework**. We observed a material uplift in performance metrics over the full history - Sharpe improved, skewness doubled, worst drawdown reduced, and correlation to broader market risk factors declined. A key driver is this model **exhibits a near-zero or negative correlation with all existing strategies**, enhancing diversification under our hierarchical risk-parity framework.

Global Quantitative and Derivatives Strategy

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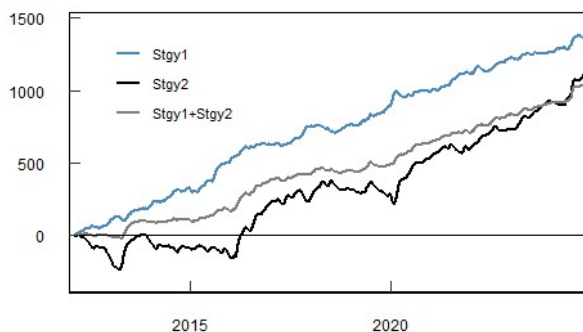
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Figure 1: Backtest performance



Source: J.P. Morgan

Figure 2: Correlation surface model exhibits low correlation with other strategies

	CM	CorrSurf	EQ	ETF	FI	FX	VarSwap	VolSurf
CM		-3%	-7%	-1%	-3%	10%	-2%	-3%
CorrSurf	-3%		2%	1%	-1%	-4%	-3%	1%
EQ	-7%	2%		7%	0%	2%	0%	3%
ETF	-1%	1%	7%		3%	4%	-3%	7%
FI	-3%	-1%	0%	3%		1%	-2%	2%
FX	10%	-4%	2%	4%	1%		2%	-3%
VarSwap	-2%	-3%	0%	-3%	-2%	2%		1%
VolSurf	-3%	1%	3%	7%	2%	-3%	1%	

Source: J.P. Morgan

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From Volatility Surface to Correlation Surface

In [previous analysis](#), we introduced a framework to predict the future dynamics of FX vol surfaces using parametric descriptions of the IVS, which leverages IV skew, smile attenuation and term structure. This methodology allows us to identify and quantify the most vulnerable vol surfaces within the FX domain, which outperforms an unstructured, statistical factor decomposition approach (e.g. PCA). The model covers 46 FX pairs across 7 expiries (2 week, 1, 2, 3, 6, 9, and 12 months) and 5 strikes (10 and 25 delta calls and puts, and ATM calls/puts). A back-test that systematically went long and short the most dislocated points in the IVS delivered robust performance. The model has produced strong performance since its launch, and has shown very low correlations with other models in our CARV portfolio, hence improving overall portfolio performance significantly (discussed [here](#)).

Leveraging our existing vol surface model, we now extend this framework to the FX implied correlation surface. While volatility and correlation are distinct market quantities, they are tightly linked through the no-arbitrage relationships embedded in FX triangles. In particular, the implied correlation between two FX pairs sharing a common pivot can be expressed as a deterministic function of their respective implied volatilities. The primary change in modeling the correlation surface lies in our simplification of the volatility surface representation: we drop the smile-related components and focus only on the term structure parameters since FX correlation swaps are typically quoted at the ATM and standard maturities. This not only reduces model complexity but also helps isolate the dynamics that matter most for correlation pricing and trading. The forward-looking correlation surface can then be used to identify pricing dislocations and trading signals.

Modeling the Correlation Surface

The data covers January 2012 to July 2025 across 41 FX correlation pairs. For each pair, we compute the implied correlation using the standard variance identity derived from triangular no-arbitrage:

$$\rho_{USD-EUR-JPY} = \frac{\sigma_{EUR/USD}^2 + \sigma_{USD/JPY}^2 - \sigma_{EUR/JPY}^2}{2 * \sigma_{EUR/USD} * \sigma_{USD/JPY}}$$

In this example, USD is the pivot currency common to EUR and JPY.

Correlation Surface Specification

To model the correlation surface, we adopt the same parametric specification $F(v_i)$ as in our prior vol model but restrict attention to ATM strikes only, which allows us to drop all coefficients related to smile and skew¹. The remaining β coefficients capture the average short-end and medium part of the term structure, where λ determines the exponential decay rate of the term structure. This results in an in-sample fit for the daily implied correlation surface that exhibits an average R^2 of 82%.

1. Δ is set as 0 for ATM options given the Garman-Kohlhagen conversion we used in our original framework $\Delta_i = (\Delta_i^{GK} - 0.5) * 100$.

$$\rho_{i,t} = \beta_0 + \beta_{1,t} \frac{1 - e^{-\lambda \tau_i}}{\lambda \tau_i} + \beta_{2,t} \left(\frac{1 - e^{-\lambda \tau_i}}{\lambda \tau_i} - e^{-\lambda \tau_i} \right)$$

$\rho_{i,t}$ denotes the implied correlation for correlation pair i at time t .
 τ_i is the time to maturity measured in years.

λ is taken as the median value by minimizing the daily sum of squared errors on the nonlinear estimation of all parameters including λ , then we fix λ and re-estimate the model for the β coefficients.

Out-of-Sample Forecasting

We continue to use the VAR model to fit the time series of β_1 and β_2 as in our original framework, and then apply those to forecast the forward implied correlation. By evaluating the predictive power across multiple horizons (Figure 3), we find that short-term predictions, led by $h=1$ month, are more stable and accurate. Among the FX correlation pairs, the USD- and EUR-triangles exhibit the highest predictability, likely due to their deeper liquidity and lower idiosyncratic volatility.

$$\rho_{i,t+h} = \beta_{0,t+h} + \beta_{1,t+h} \frac{1 - e^{-\lambda \tau_i}}{\lambda \tau_i} + \beta_{2,t+h} \left(\frac{1 - e^{-\lambda \tau_i}}{\lambda \tau_i} - e^{-\lambda \tau_i} \right)$$

Figure 3: Out-of-sample forecast accuracy

Constructed forecasts of the implied correlation surface sby adjusting h from 1 month to 1 year

Name \ Days ahead (h)	RMSE					R2				
	1M	3M	6M	9M	1Y	1M	3M	6M	9M	1Y
EUR-USD-CHF	0.005	0.071	0.024	0.028	0.024	61.04	51.13	38.92	90.85	50.23
EUR-CAD-GBP	0.012	0.016	0.034	0.005	0.029	62.54	0.67	47.35	80.13	74.95
USD-EUR-CHF	0.012	0.018	0.043	0.074	0.087	77.53	17.49	2.17	2.47	61.38
EUR-CAD-NZD	0.013	0.019	0.041	0.036	0.071	64.31	0.00	61.29	38.53	28.18
USD-EUR-NOK	0.025	0.104	0.054	0.048	0.061	66.45	64.40	6.41	78.91	5.86
EUR-JPY-CHF	0.025	0.046	0.037	0.041	0.010	37.87	60.74	17.79	11.84	6.13
EUR-CAD-AUD	0.027	0.054	0.074	0.052	0.106	88.28	46.60	50.13	49.57	38.32
EUR-USD-GBP	0.027	0.015	0.073	0.047	0.103	6.76	39.63	0.84	70.42	4.55
USD-JPY-CAD	0.028	0.024	0.024	0.025	0.061	41.90	3.55	58.57	92.68	0.19
EUR-USD-NZD	0.030	0.038	0.099	0.129	0.162	38.55	4.68	2.68	4.01	39.59
USD-CAD-AUD	0.030	0.028	0.058	0.060	0.066	33.35	0.79	18.76	28.48	54.82
EUR-USD-NOK	0.040	0.010	0.156	0.211	0.195	11.44	63.66	13.06	23.40	0.50
EUR-USD-AUD	0.043	0.062	0.127	0.141	0.197	64.19	0.96	5.23	14.15	62.22
USD-EUR-AUD	0.044	0.138	0.093	0.082	0.123	92.22	81.25	39.86	53.45	60.49
USD-CAD-NZD	0.049	0.050	0.072	0.078	0.092	58.44	0.16	31.71	31.06	87.41
EUR-USD-CAD	0.050	0.101	0.081	0.057	0.071	93.14	68.82	51.68	73.25	0.01
USD-JPY-AUD	0.050	0.050	0.050	0.067	0.089	88.42	60.58	77.01	59.75	1.61
EUR-USD-CNH	0.051	0.050	0.084	0.066	0.122	35.05	1.48	24.90	60.52	9.06
EUR-USD-SGD	0.051	0.054	0.084	0.076	0.067	9.78	0.72	1.93	4.62	12.66
USD-EUR-KRW	0.053	0.073	0.095	0.112	0.102	8.06	13.27	5.82	9.55	0.00
USD-EUR-NZD	0.054	0.124	0.088	0.098	0.125	89.55	63.77	56.18	50.50	38.38
EUR-JPY-SGD	0.055	0.045	0.042	0.020	0.026	48.59	3.03	27.02	1.39	30.51
EUR-USD-KRW	0.056	0.106	0.111	0.129	0.184	80.68	60.02	85.73	36.40	37.03
USD-JPY-NZD	0.058	0.053	0.047	0.067	0.086	71.81	58.56	83.84	32.25	1.40
USD-JPY-GBP	0.058	0.067	0.053	0.058	0.094	48.40	9.36	62.85	40.56	15.60
USD-EUR-CAD	0.062	0.089	0.077	0.068	0.094	49.97	57.55	10.85	63.74	1.04
EUR-JPY-NZD	0.063	0.061	0.087	0.078	0.057	20.98	41.31	29.79	50.93	90.89
EUR-JPY-AUD	0.073	0.077	0.094	0.083	0.070	26.74	39.29	49.39	54.73	95.00
USD-JPY-CHF	0.076	0.087	0.084	0.077	0.074	51.53	0.49	45.86	71.76	3.26
USD-EUR-JPY	0.076	0.075	0.070	0.053	0.068	50.93	1.32	38.93	65.82	11.14
USD-CAD-GBP	0.082	0.103	0.108	0.111	0.137	70.52	60.22	2.64	5.68	61.95
EUR-JPY-CAD	0.090	0.108	0.120	0.115	0.114	28.48	10.60	1.19	30.15	69.33
EUR-JPY-GBP	0.095	0.154	0.143	0.135	0.162	81.02	31.33	24.15	61.99	79.77
USD-AUD-NZD	0.099	0.078	0.078	0.067	0.070	56.77	52.68	2.54	47.99	86.22
USD-JPY-KRW	0.101	0.128	0.169	0.126	0.190	94.44	92.06	17.47	58.53	31.14
USD-JPY-SGD	0.106	0.064	0.142	0.170	0.190	82.49	92.97	31.48	0.23	31.49
EUR-AUD-NZD	0.108	0.074	0.060	0.047	0.039	59.59	33.58	38.04	7.76	75.39
USD-EUR-GBP	0.123	0.191	0.185	0.165	0.182	80.17	7.13	1.48	18.62	0.06
EUR-USD-JPY	0.130	0.166	0.228	0.161	0.233	58.49	26.82	19.76	54.03	33.61
USD-EUR-SGD	0.148	0.100	0.120	0.107	0.159	0.00	17.11	27.35	78.51	41.23
USD-EUR-CNH	0.159	0.178	0.172	0.144	0.164	2.76	4.24	14.45	30.67	2.05

Source: J.P. Morgan

Trading the Implied Correlation Forecasts

We evaluate the performance of our implied correlation forecasts by constructing a systematic trading strategy on FX correlation swaps. Specifically, we implement and compare three strategies that take directional positions based on model signals versus market pricing:

Strategy 1: Market-Implied vs. Model-Implied Correlation

For each currency and for a given forecast horizon h , on any given day, we rank all the correlation pairs based on the deviation between the h -day-ahead implied correlation forecast and the current implied correlation level in ascending order (i.e. the largest forecast - current spread has a high rank). Therefore, a low (high) rank suggests a potential sell (buy) signal. The idea is that if our model accurately captures forward-looking dynamics, **any persistent dislocation between model-implied and market-implied correlations reflects an opportunity to trade**.

Next, we define P^- and P^+ to be the set of top p and bottom p ranks, and AVG_{low} and AVG_{high} to be the average of the deviations for each set, respectively. The more negative the AVG_{low} is, the stronger the sell signal for the options in set P^- ; conversely, the more positive the AVG_{high} is, the stronger the buy signal for the options in set P^+ .

On each trading day and for each correlation pair, we implement a **long-only or short-only strategy** based on the following rules:

- When $AVG_{low} > 0$, all options have a higher forecasted implied correlation than the current level, indicating there are no sell signals and thus we go **long the correlation swap** that screens the cheapest (those under P^+).
- When $AVG_{high} < 0$, all options have a lower forecasted implied correlation than the current level, indicating there are no buy signals and thus we go **short the correlation swap** that screens the richest (those under P^-).
- When $AVG_{low} < 0$ and $AVG_{high} > 0$,
 - If $|AVG_{low}| > |AVG_{high}|$, there are more deviations in P^+ , thus we go **long the correlation swap** in P^+ .
 - If $|AVG_{low}| < |AVG_{high}|$, there are more deviations in P^- , thus we go **short the correlation swap** in P^- .

Once the trading universe and the positions are determined, the weight of each option is scaled by the degree of the deviation. We take $h = 1$ month and $p = 5$ ranks as an examples. Each position is held for 1 month.

Strategy 2: Market Implied vs. Realized Correlation (Benchmark)

Instead of comparing the market-model implied correlation, we take the difference between the current implied correlation level vs. the trailing realized correlation. This doesn't use the model's predictions and we refer to this as the benchmark strategy. We then implement the same ranking and deviation-based weighting process (i.e. repeat the same exercise as in strategy 1).

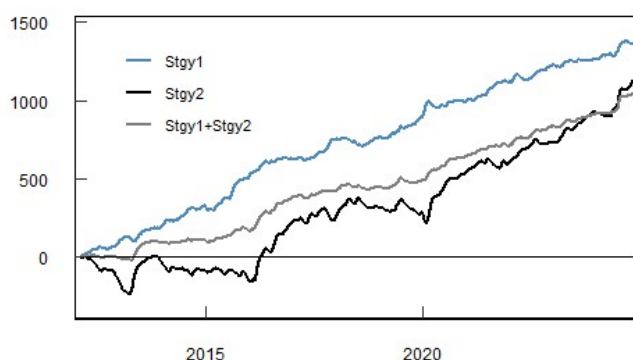
Strategy 3: Strategy 1+ Strategy 2 Hybrid

To refine the signal quality further, we construct a hybrid strategy that takes trades **only when both Strategy 1 and Strategy 2 independently generate a signal in the same direction**. While this reduces the number of positions, it ensures higher conviction and avoids potentially noisy or contradictory signals.

The strategy performance is shown in the full sample and the post-2020 period (Figure 4 and Figure 5):

- **Strategy 1 consistently outperforms Strategy 2** - It delivers a higher Sharpe ratio (1.44 vs. 0.77) over the full history of the study, and shows even stronger performance post-2020 (1.74 vs. 1.24), indicating that model-implied correlation provides a more robust and timely signal than relying on realized correlation alone.
- **Strategy 3 delivers the highest Sharpe ratio post 2020 (1.97)** despite fewer positions, suggesting that combining both signals enhances the risk-adjusted return. It also exhibits lower drawdown and higher positive skewness, reinforcing its quality as a high conviction approach.

Figure 4: Strategy performance



Source: J.P. Morgan

The strategy collects 10 corr pts per month if no TC included. Assuming TC is would half the PnL, it still shows decent results.

Source: J.P. Morgan

Figure 5: Strategy metrics

	Full History			Post 2020		
	Strategy 1	Strategy 2	Strategy 1+2	Strategy 1	Strategy 2	Strategy 1+2
No. of position /day	203	197	82	210	200	84
Annualized PNL	4.93	4.05	3.77	8.27	4.18	5.25
Annualized Vol	3.43	5.25	2.69	4.75	3.38	2.67
Sharpe	1.44	0.77	1.40	1.74	1.24	1.97
Skewness	0.28	0.75	0.98	1.23	0.27	1.71
WDD	-9.24	-13.06	-6.11	-10.44	-8.00	-6.11
Down Dev	1.96	2.67	1.36	2.01	1.91	1.05
Sortino	2.51	1.52	2.77	4.12	2.19	4.99

Source: J.P. Morgan

Contribution to Portfolio-Level Performance

Beyond its standalone performance, this model meaningfully **enhances our cross-asset portfolio** when integrated within our broader multi-strategy framework (discussed [here](#)).

Recall that our aggregate portfolio combines 9 systematic strategies across Equities, FX, Rates, and Commodities using a hierarchical risk parity approach. This method clusters risk by currency and asset class, ensuring diversified and balanced exposure while avoiding concentration in any one macro driver.

After incorporating our correlation strategy into the portfolio, we observed **a material uplift in performance metrics** over the full history (Figure 7) - **Sharpe ratio improved significantly, skewness doubled, worst drawdown reduced, and correlation to broader market risk factors declined**.

A key driver behind these improvements is the **extremely low correlation profile of this model relative to our existing models** (Figure 6) - the correlation with any

individual strategy in the portfolio is between -4% and 2%, and it shows negative correlation with the FX gamma model and just +1% correlation with the FX vol surface model. This emphasizes the complementary nature of the correlation signal.

Figure 6: Correlation Map

	CM	CorrSurf	EQ	ETF	FI	FX	VarSwap	VolSurf
CM		-3%	-7%	-1%	-3%	10%	-2%	-3%
CorrSurf	-3%		2%	1%	-1%	-4%	-3%	1%
EQ	-7%	2%		7%	0%	2%	0%	3%
ETF	-1%	1%	7%		3%	4%	-3%	7%
FI	-3%	-1%	0%	3%		1%	-2%	2%
FX	10%	-4%	2%	4%	1%		2%	-3%
VarSwap	-2%	-3%	0%	-3%	-2%	2%		1%
VolSurf	-3%	1%	3%	7%	2%	-3%	1%	

Source: J.P. Morgan

Figure 7: Portfolio Stats

	10Y History	5Y History	YTD	Last quarter
Annual PnL	9.33	3.65	7.29	2.16
Sharpe Ratio	6.49	4.07	6.43	2.69
Skewness	0.79	0.66	1.18	1.03
Worst Drawdown	-1.42	-0.39	-0.70	-0.38
Correlation with S&P	0.01	0.09	0.00	0.12

Source: J.P. Morgan

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